

CHASMBrain: Coarse-to-fine Hierarchical Architecture with Sequential Mamba for Brain Reconstruction

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Abstract. Understanding the relationship between deep visual representations and the human visual system is a fundamental challenge in computational neuroscience. While modern vision models achieve strong performance in image recognition, their correspondence with the hierarchical organization of the human visual cortex has not been thoroughly investigated. In this study, we propose CHASMBrain, a novel hierarchical two-stage framework for image-to-fMRI encoding. Our architecture leverages a dual-stream Mamba design to explicitly separate and process global semantic tokens and local spatial patches, better matching the functional organization of the visual cortex. A coarse-to-fine strategy is employed: Stage 1 predicts denoised ROI-level activations, while Stage 2 refines these coarse responses into full voxel-level predictions using a Mamba-VAE. Experiments on the large-scale Natural Scenes Dataset (NSD) demonstrate that our method achieves a Pearson correlation of 0.429 and an MSE of 0.261, significantly outperforming baseline approaches. Beyond improving prediction performance, our findings provide empirical evidence that self-supervised features from DINOv2 align with the cortical hierarchy, offering an interpretable and biologically-inspired framework for neural visual encoding.

Keywords: Image-to-fMRI · Computational Neuroscience · Vision Transformer · Brain Encoding

1 Introduction

Vision is the primary sensory channel through which humans acquire information about the external world. Even a single second of natural visual input carries a staggering amount of information. To understand how the brain parses this input, computational neuroscience relies on visual encoding models that describe neural responses as functions of stimulus features. Functional magnetic resonance imaging (fMRI) is central to this effort: by measuring blood-oxygen-level-dependent signals, fMRI provides a non-invasive way to map visual stimuli to localized responses in the human visual cortex.

Historically, encoding models depended on hand-crafted features. Early work used mathematical constructs such as Gabor filters to approximate sensitivity to edges, orientations, and contrast. As naturalistic vision benchmarks became

more challenging, the field shifted from manual descriptors to learning-based approaches powered by deep neural networks, which extract rich data-driven representations directly from images.

This transition also revealed a strong correspondence between artificial hierarchies and biological vision. Prior studies show that layers of deep neural networks (DNNs) map onto successive processing stages in human visual cortex [10]. Early convolutional layers are predictive of activity in low-level areas such as V1 and V2 [17], while deeper layers better align with higher-level regions, including inferior temporal (IT) cortex and fusiform face area (FFA), that support complex object recognition [8]. These findings suggest shared computational principles across artificial and biological systems.

To better capture this organization, recent models explicitly separate local and global representations. Early visual cortex is retinotopic, making it especially sensitive to local, patch-level structure. Higher-level regions encode more holistic semantics that reflect object identity and behavioral relevance, beyond low-level geometry [3]. Disentangling local spatial pooling (“where”) from global feature tuning (“what”) has therefore become a key design principle, as shown by frameworks such as banded ridge regression and targeted transformer queries [1, 16].

This separability is consistent with evidence for structural similarity between deep visual representations and the human visual system. From striate to extrastriate cortex, representational complexity increases along a spatial hierarchy [5]. Deep models exhibit a comparable layer-wise progression [14], often described as a form of hierarchical isomorphism between model stages and anatomical pathways spanning ventral and dorsal streams [7, 18].

Building on these insights, this paper presents an interpretable, biologically inspired dual-stream framework for neural visual encoding. While prior deep models often achieve high predictive accuracy, they can remain difficult to interpret. The proposed framework incorporates cortical priors by separating spatial location parameters from feature-weight parameters, enabling direct inspection of receptive-field structure [7, 16]. It further mirrors neuroanatomical specialization by decoupling spatial detail from semantic context, following evidence from brain-like hierarchical processing [6]. This design aims to improve both predictive performance and scientific interpretability.

To bridge the gap between artificial visual representations and human neural activity as discussed, the core contributions of this work are summarized as follows:

- Propose CHASMBrain, a hierarchical two-stage framework for image-to-fMRI reconstruction that achieves good results. Our model yields a Pearson correlation of 0.429 and an MSE of 0.261.
- Develop a sequential coarse-to-fine learning mechanism specifically designed to overcome the challenge of high intrinsic noise in fMRI signals. By separating robust semantic alignment at the ROI level from fine-grained voxel-level refinement.
- Provide empirical evidence of biological alignment by demonstrating that the self-supervised features of DINOv2 [11] closely mirror the hierarchical pro-

cessing of the human visual system. Through regional analysis, we confirm that our dual-stream mamba architecture effectively captures the structural similarity between deep visual representations and the functional organization of the visual cortex.

2 Proposed Method

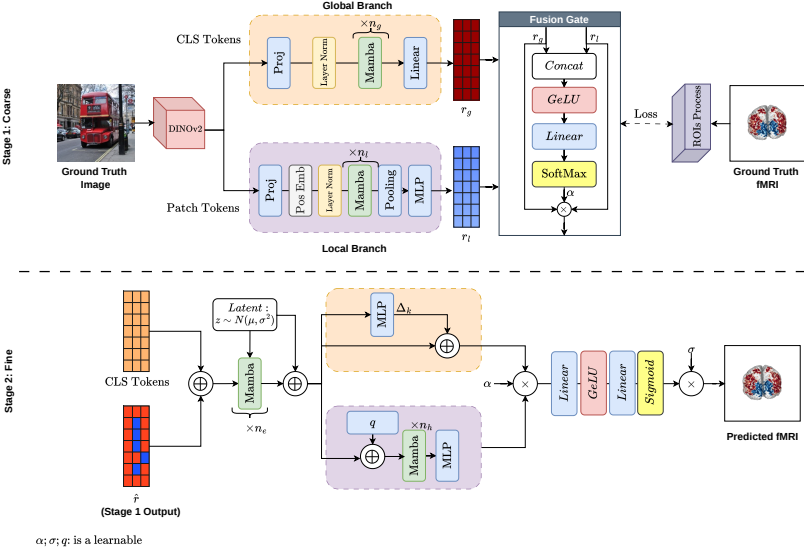


Fig. 1: Overview of the proposed hierarchical two-stage framework for image-to-fMRI reconstruction. **Stage 1 (Coarse):** A dual-stream Mamba architecture processes global (CLS token) and local (patch tokens) visual features separately, then fuses them via an adaptive gate to predict ROI-level activations. **Stage 2 (Fine):** A Mamba-VAE architecture takes the visual features and ROI predictions as input, encodes them through Mamba blocks, samples from a learned latent space, and decodes to full voxel-level fMRI predictions through parallel global and local branches.

We propose a hierarchical two-stage framework for image-to-fMRI reconstruction, as shown in Figure 1. Stage 1 first predicts denoised responses at the ROI (Region of Interest) level, and Stage 2 then refines these coarse responses into full voxel-level predictions. This design separates robust semantic alignment from fine-grained signal recovery, which makes the overall learning process more stable. Crucially, Stage 1's dual-stream architecture extracts and encodes local spatial details into ROI representations, allowing Stage 2 to focus on refinement using these encoded structures rather than re-extracting features from raw patches.

2.1 Stage 1: Dual-Stream Mamba for Coarse Prediction

Stage 1 learns ROI-level neural responses from visual inputs by combining complementary cues: global semantics and local spatial details. To this end, we use a dual-stream architecture where each stream specializes in one type of visual representation and both streams are fused adaptively.

Visual Feature Extraction Given an input image I , we extract visual features using a pretrained DINOv2 encoder [11]. The encoder outputs two representations:

- **CLS token**: a single vector summarizing global image semantics.
- **Patch tokens**: a sequence of vectors preserving local spatial structure.

We choose DINOv2 because it is pretrained via self-supervised learning, which is effective for capturing fine visual details and structural patterns. This property is well aligned with our hypothesis that both detailed local structure and global visual semantics are critical for accurate image-to-fMRI reconstruction.

ROI Aggregation for Noise Reduction Directly regressing all voxels (e.g., 15,724 dimensions) is difficult because fMRI signals are high-dimensional and noisy, with variance from physiological artifacts, scanner noise, and trial-level fluctuations. Therefore, Stage 1 predicts ROI-level targets as an intermediate denoising step.

We partition voxels into K non-overlapping clusters, each containing V spatially adjacent voxels. Compared with anatomical parcellations such as Schaefer [13], our sequential clustering provides flexible granularity tailored to the target voxel space.

For each cluster, instead of averaging all voxels, we aggregate only the top- $p\%$ activations:

$$r_k = \frac{1}{|S_k|} \sum_{i \in S_k} v_i \quad \text{where} \quad S_k = \text{TopK}(\{v_i\}_{i \in C_k}, \lfloor p \cdot V \rfloor) \quad (1)$$

where r_k is the aggregated value of ROI k , C_k is the voxel index set of cluster k , and TopK returns the highest-activation subset. This strategy preserves informative peaks while reducing low-activation noise; it is especially useful after z-score normalization, where naive mean aggregation can collapse toward zero.

Global Branch The Global branch models high-level semantics from the CLS token. The projected CLS features are treated as a sequence across the Mamba block [4] depth, allowing selective state-space modeling to filter task-irrelevant dimensions progressively

We adopt Mamba primarily because its selective state-space mechanism can focus on task-relevant information while suppressing irrelevant signals, which aligns well with brain activity prediction.

Local Branch The Local branch models fine-grained structure from patch tokens. We first project patch tokens, add learnable positional embeddings, and apply normalization. The sequence is then processed by a deeper stack of Mamba blocks than in the Global branch, reflecting the larger token sequence (256 patches). Finally, mean pooling and a two-layer GELU MLP produce ROI-level predictions.

Adaptive Fusion Gate To combine the two streams, we use a learnable fusion gate rather than fixed coefficients. The gate takes concatenated global/local features, applies a two-layer GELU MLP, and outputs softmax-based fusion weights. The final ROI prediction is

$$\hat{\mathbf{r}} = \alpha \cdot \mathbf{r}_g + (1 - \alpha) \cdot \mathbf{r}_l, \quad (2)$$

where $\mathbf{r}_g$ and $\mathbf{r}_l$ are branch outputs and $\alpha \in [0, 1]$ is learned per input. This mechanism lets the model emphasize global context for simple scenes and local details for complex scenes.

Contrastive Alignment Beyond regression, we encourage representation-level alignment between image features and brain responses via a CLIP-style contrastive objective [12]. Specifically, fused visual features and ground-truth ROI activations are projected into a shared embedding space using separate two-layer GELU projection heads with layer normalization. An InfoNCE loss then pulls matched image–brain pairs together while pushing mismatched pairs apart.

Training Objective The Stage 1 objective combines pointwise regression, correlation alignment, and contrastive alignment:

$$\mathcal{L}_{stage1} = \mathcal{L}_{MSE} + \gamma \cdot \mathcal{L}_{Pearson} + \lambda \cdot \mathcal{L}_{InfoNCE}. \quad (3)$$

Here, $\mathcal{L}_{MSE}$ minimizes amplitude error between predicted and ground-truth ROI activations, $\mathcal{L}_{Pearson}$ encourages high correlation between their response patterns, and $\mathcal{L}_{InfoNCE}$ improves cross-modal discriminability. Hyperparameters γ and λ balance these three terms.

2.2 Stage 2: Mamba-VAE for Fine Prediction

Stage 2 maps coarse ROI predictions to full voxel resolution. We design this stage as a Mamba-VAE to jointly model structured dependencies and trial-to-trial variability in fMRI responses.

Input Fusion Stage 2 receives two inputs: visual features from DINOv2 and ROI predictions from Stage 1 ($\hat{\mathbf{r}}$). We first project visual features (CLS token concatenated with pooled patch features) into a common embedding space. Notably, we use mean-pooled patch features rather than individual patch tokens, as

Stage 1’s Local branch has already extracted and encoded fine-grained spatial details into the ROI representations. This avoids redundant feature extraction and reduces computational overhead while maintaining spatial specificity through the ROI structure. Each ROI value is then tokenized and augmented with learnable positional embeddings to preserve region order. Concatenating visual and ROI tokens forms a unified sequence encoding both stimulus content and coarse neural priors.

Mamba Encoder A stack of Mamba blocks encodes the fused sequence and learns interactions between visual evidence and predicted neural structure. This encoder produces latent token representations that capture global image–brain dependencies before voxel-level decoding.

Variational Latent Space Because fMRI responses vary even for repeated presentations of the same stimulus, we model uncertainty with a variational latent space. After sequence encoding, pooled features are passed to a VAE head:

$$\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2 = \text{VAEEncoder}(\text{Pool}(\mathbf{h})). \quad (4)$$

During training, we sample

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (5)$$

and inject $\mathbf{z}$ back into token representations. During inference, we use $\boldsymbol{\mu}$ for deterministic prediction.

Feature Split for Dual Decoding After latent injection, we obtain a shared token sequence $\tilde{\mathbf{H}} \in \mathbb{R}^{(K+1) \times d}$, including one visual token and K ROI tokens. We then split features by token role (not by separate encoders): the full sequence is routed to a holistic decoder, while ROI tokens are selected and routed to an ROI-wise residual decoder. Formally,

$$\tilde{\mathbf{H}}_{global} = \tilde{\mathbf{H}}, \quad \tilde{\mathbf{H}}_{local} = \tilde{\mathbf{H}}_{2:K+1}. \quad (6)$$

This token-level split allows both decoders to share the same encoded context while specializing in complementary prediction behaviors.

Holistic Voxel Decoder The Holistic Voxel Decoder predicts the full voxel vector jointly from the global context. We append learnable query tokens $\mathbf{q} \in \mathbb{R}^{M \times d}$ to the latent-injected sequence, process them with additional Mamba blocks, and map the resulting query features to voxel outputs. This branch captures long-range and cross-ROI dependencies.

ROI Residual Decoder The ROI Residual Decoder predicts ROI-wise voxel details through residual learning. For each ROI token, a small MLP estimates voxel residuals relative to the Stage 1 coarse value:

$$\mathbf{v}_k = r_k + \Delta_k, \quad (7)$$

where r_k is the coarse ROI prediction and Δ_k is the local residual. This formulation uses coarse predictions as priors and focuses local modeling capacity on fine corrections.

Prediction Fusion and Amplitude Recovery The fused voxel prediction combines holistic and residual outputs:

$$\mathbf{v}_{fused} = \alpha \cdot \mathbf{v}_{global} + (1 - \alpha) \cdot \mathbf{v}_{local}, \quad (8)$$

where α is a single global learnable parameter, in contrast to Stage 1’s input-dependent adaptive gate. This simpler fusion is sufficient because both decoders read from the same shared representation and differ only in decoding granularity (holistic vs. ROI-residual).

To recover appropriate response amplitudes, we apply a learnable scale-recovery layer conditioned on visual features:

$$s = \text{Sigmoid}(\text{MLP}(\mathbf{h}_{vis})) \cdot \sigma, \quad (9)$$

where $\mathbf{h}_{vis}$ is the visual token from the encoded sequence and σ is a learnable scale bound. The final prediction is $\hat{\mathbf{v}} = s \cdot \mathbf{v}_{fused}$, allowing the model to adjust output magnitude based on stimulus complexity.

Training Objective The Stage 2 objective balances voxel reconstruction and latent regularization:

$$\mathcal{L}_{stage2} = \mathcal{L}_{MSE} + \beta \cdot \mathcal{L}_{KL}, \quad (10)$$

where $\mathcal{L}_{MSE}$ is voxel-level reconstruction loss and $\mathcal{L}_{KL} = D_{KL}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$ regularizes the posterior toward a standard Gaussian prior. The coefficient β controls this trade-off.

3 Experiments

3.1 Experimental Setup

Dataset and Features. We evaluate our method on the Natural Scenes Dataset (NSD) [2]. Following standard protocols in Mindsimulator [19] Mindeye2 [15], we use data from subjects 1, 2, 5, and 7. For each subject, we allocate 9,000 image-fMRI pairs for training and 1,000 for testing, focusing on visual cortex regions which yield approximately 15,000 voxels per subject. Visual representations are

extracted using a pre-trained DINOv2-ViT-B/14 [11], producing a 768-D global CLS token and 256 local patch tokens.

Implementation Details. CHASM-Brain is implemented in PyTorch and trained on a single RTX 4090 GPU. Stage 1 is trained for 150 epochs (batch size 32, AdamW, learning rate 10^{-4} , InfoNCE weight $\lambda = 0.1$). Stage 2 is trained for 150 epochs (batch size 16, learning rate 10^{-4} , KL weight $\beta = 0.001$). For region-of-interest (ROI) clustering, we set $K = 30$ voxels per cluster with top-70% aggregation.

Metrics and Baselines. We evaluate predictive performance using Mean Squared Error (MSE, $\downarrow$) and voxel-wise Pearson Correlation ($\uparrow$). We compare against two state-of-the-art visual encoding models: SynBrain [9] and MindSimulator [19]. Following the NSD analysis framework [2], measurement fidelity is quantified by the noise ceiling estimated from single-trial GLM betas (the refined “b3” pipeline with voxel-wise HRF estimation, nuisance regression, and fractional ridge regression). The reported average predictable variance in visual cortex is approximately 36%, which corresponds to a theoretical maximum of Pearson $r \approx 0.60$. Therefore, correlations substantially exceeding 0.60 are likely physiologically implausible under this benchmark and should be interpreted as potentially non-natural or protocol-inconsistent estimates.

3.2 Quantitative Results

Table 1 presents the voxel-level fMRI prediction performance. Our proposed CHASM-Brain achieves state-of-the-art results, yielding an MSE of 0.261 and a Pearson correlation of 0.429. This represents a substantial 36.8% reduction in MSE and a 33.2% relative correlation improvement over MindSimulator.

Notably, we reproduce SynBrain utilizing subject-wise z-score normalization—a rigorous standard necessary for downstream generative tasks [19]—rather than their original global constant scaling (/2000). Under this evaluation protocol that better reflects the biological noise ceiling, CHASM-Brain demonstrates clear superiority. This strong performance validates our core design: explicit disentanglement of global and local visual features coupled with a coarse-to-fine generative refinement strategy effectively bridges the semantic-spatial gap in neural encoding.

Table 1: Comparison with state-of-the-art methods on voxel-level fMRI prediction. All results are evaluated on the same test set. *SynBrain is reproduced using standardized subject-wise z-score normalization for fair evaluation.

Model	Venue / Year	MSE $\downarrow$	Pearson $\uparrow$
SynBrain* [9]	NeurIPS 2025	0.4533	0.3376
MindSimulator [19]	ICLR 2025	0.429	0.322
CHASM-Brain (Ours)	-	0.261	0.429

3.3 Qualitative Results

Predicted fMRI Maps and Visual Reconstruction. To intuitively evaluate our predictions, we analyze both the spatial fMRI activation patterns and their downstream applicability for visual reconstruction (Figure 2).

First, mapping the predicted signals back to the cortical surface reveals that CHASM-Brain produces highly accurate spatial activation topologies. Compared to baselines that often yield overly smoothed or scattered noise, our hierarchical approach preserves sharp regional boundaries and localized peaks that closely mirror the ground-truth neural responses.

Second, we feed these predicted fMRI signals into an fmri-to-image decoder (MindEye2) to reconstruct visual stimuli. It is crucial to acknowledge that the visual fidelity of these downstream 2D reconstructions is heavily bottlenecked by the inherent capabilities and generative priors of the chosen decoder. Even when CHASM-Brain yields highly accurate spatial activation topologies, the final reconstructed images remain highly sensitive to MindEye2’s decoding mechanisms and potential hallucination tendencies. This dependency is evidenced by the fact that even ground-truth fMRI signals fail to perfectly restore original object attributes, establishing a strict visual upper bound. Nevertheless, within these constraints, CHASM-Brain demonstrates robust semantic retention. For instance, our method successfully reconstructs coherent structures of rigid objects (e.g., the red double-decker bus, airplane) significantly better than MindSimulator. Occasional semantic drifts (e.g., misinterpreting a snow-skiing scene as a water sport) highlight that while global categories are strongly captured, resolving highly specific contextual details from simulated fMRI predictions remains challenged by the decoder’s generative biases.

Dual-Stream Biological Alignment. To validate our biological motivation, we map the predictive correlations of the Local (patch) and Global (CLS) branches onto the visual cortex hierarchy (Figure 3).

The Local branch (blue) exhibits strong correlations with early visual areas (V1, V2), which are retinotopically organized to process low-level spatial frequencies and edges. Conversely, the Global branch (red) heavily activates higher-order semantic regions across the lateral and ventral streams, areas responsible for holistic object recognition. This spatial dichotomy provides compelling empirical evidence that our explicitly decoupled dual-stream design effectively mimics the hierarchical processing mechanisms of the human visual system.

3.4 Ablation Studies

Architecture and Refinement Strategy. Table 2 dissects our architectural components. For Stage 1 coarse prediction (Table 2a), the proposed Dual-Mamba (Pearson: 0.585) significantly outperforms both a standard Transformer (0.452) and a single-stream Mamba (0.481). This confirms that explicitly branching global semantics and local spatial details is crucial for aligning with cortical representations.



Fig. 2: Qualitative comparison of predicted fMRI activations and downstream image reconstructions. CHASM-Brain accurately preserves spatial topographic patterns and enables the downstream decoder (MindEyeV2) to reconstruct visual semantics that closely align with the original stimuli.

For Stage 2 voxel refinement (Table 2b), our VAE-based stochastic refinement achieves the best synthesis (MSE: 0.261). Interestingly, diffusion-based refinement fails catastrophically (MSE: 1.973). We attribute this to the high intrinsic physiological noise of fMRI signals: the forward noise-injection process of diffusion models conflates artificial Gaussian noise with biological noise, severely degrading the signal. In contrast, our VAE smoothly models trial-to-trial voxel variations via KL regularization without explicit noise injection.

Clustering Granularity. Table 3 demonstrates the trade-off inherent in ROI clustering. Coarse clustering (100 voxels/cluster) maximizes Stage 1 correlation (0.605) through aggressive denoising but bottlenecks Stage 2 refinement (MSE drops to 0.561) due to the highly ill-posed one-to-many voxel mapping. Conversely, fine clustering (10 voxels/cluster) preserves spatial resolution for Stage 2 but suffers in Stage 1 due to insufficient noise averaging. Our empirical choice of 30 voxels/cluster strikes the optimal balance. Furthermore, the anatomically predefined Schaefer parcellation [13] performs poorly, proving that flexible, data-driven sequential clustering is necessary to capture task-specific visual cortex activations.

3.5 Discussion and Limitations

While CHASM-Brain establishes state-of-the-art visual encoding performance, certain limitations remain. First, its predictive upper bound is constrained by the inherent physiological noise of fMRI signals. Second, our pipeline relies on a frozen DINOv2 encoder, meaning the semantic representation is biased toward

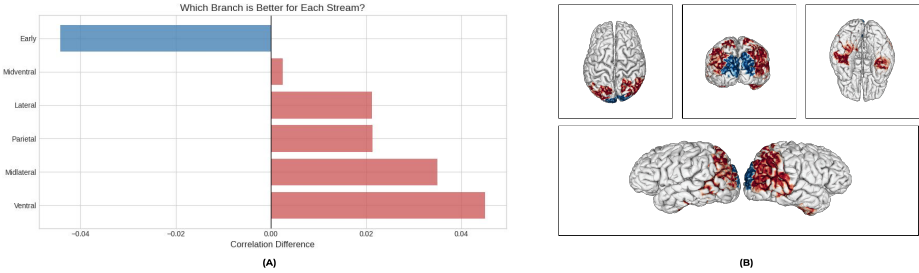


Fig. 3: Spatial analysis of dual-stream correlations. **Left:** Early visual areas (V1/V2) are primarily driven by the Local branch (blue), while higher-order semantic areas are driven by the Global branch (red). **Right:** 3D visualization showing distinct functional alignments.

Table 2: Ablation studies evaluating the Stage 1 coarse architecture (a) and the Stage 2 voxel-level refinement strategy (b).

(a) Stage 1: Architecture Design

Architecture	MSE ↓ Pearson ↑	
Transformer	0.213	0.452
Mamba (Single)	0.156	0.481
Dual-Mamba (Ours)	0.141	0.585

(b) Stage 2: Refinement Strategy

Method	MSE ↓ Pearson ↑	
MLP (Baseline)	0.535	0.391
w-Dual only	0.399	0.406
w-Dual + Diff	1.973	0.259
w-Dual + VAE (Ours)	0.261	0.429

the pre-training dataset (e.g., static, object-centric images) and may lack the temporal dynamics intrinsic to natural human vision. Finally, downstream qualitative evaluations are inevitably bottlenecked by the hallucination tendencies of the fMRI-to-image model.

4 Conclusion

In this paper, we introduced CHASMBrain, a novel coarse-to-fine hierarchical architecture for brain reconstruction. By incorporating a sequential Mamba-based dual-stream design, our model effectively bridges the gap between deep visual features and the hierarchical processing mechanism of the human visual system. The proposed two-stage approach—moving from denoised ROI predictions to

Table 3: Impact of clustering granularity on both prediction stages.

Clustering Strategy	Stage 1 (Coarse)		Stage 2 (Fine)	
	MSE ↓	Pearson ↑	MSE ↓	Pearson ↑
10 voxels/cluster (1500 clusters)	0.242	0.543	0.332	0.375
30 voxels/cluster (500 clusters)	0.141	0.585	0.261	0.429
100 voxels/cluster (150 clusters)	0.123	0.605	0.561	0.370
Schaefer Atlas [13]	0.486	0.338	0.577	0.236

fine-grained voxel-level refinement—successfully addresses the challenge of high intrinsic noise in fMRI signals while maintaining spatial and semantic specificity.

Our experimental results on the NSD dataset demonstrate state-of-the-art performance, yielding a Pearson correlation of 0.429 and a substantial reduction in MSE to 0.261. Crucially, our spatial analysis confirms a strong biological alignment: the local branch accurately maps to early visual regions, whereas the global branch captures higher-order semantic processing. While the current model is limited to static images and subject-specific mapping, it provides a robust foundation for future research. Moving forward, we intend to expand this research in two primary directions. First, we aim to conduct a more detailed investigation into the fine-grained alignment between deep learning architectures and brain regions, exploring how specific model components correspond to sub-regional functional specializations within the cortex. Second, we plan to evolve CHASMBrain into a multi-modal encoding system, extending its capabilities beyond static images to include diverse sensory channels such as audio, video, and language. We believe this path will lead to a more profound understanding of how the human brain integrates complex, multi-sensory information.

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